

ITUS Feedback Module – Excel Upload API Contract

API Specification for file injection into the ITUS Feedback Module

Version: 1.0

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Base URL: `{VITE_API_URL}`

Auth: Bearer JWT

Content-Type: multipart/form-data

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1. Overview & Common Conventions

Property	Value
Protocol	HTTPS
Base URL	<code>{VITE_API_URL}</code> (configured per environment)
Request body (upload)	<code>multipart/form-data</code>
Response body	<code>application/json</code>
Supported file types	<code>.xlsx</code> , <code>.xls</code>
Maximum file size	10 MB (enforced by Spring <code>MaxUploadSizeExceededException</code>)
Encoding	UTF-8
HTTP success code	200 OK – returned even when row-level errors are present; inspect <code>errorCount</code> in the body

Important: A 200 OK response does *not* guarantee data was saved. Always check `errorCount` in the response body. Records are only persisted when `errorCount` `===` 0.

2. Authentication

All upload endpoints require a valid JWT Bearer token issued by the ITUS authentication service.

REQUEST HEADER

Header	Required	Description
<code>Authorization</code>	Required	<code>Bearer <access_token></code> — JWT obtained from the login endpoint

Missing or expired tokens return **HTTP 401 Unauthorized**. Insufficient role/jurisdiction permissions return **HTTP 403 Forbidden**.

3. Shared Data Models

3.1 UploadResponse (Success Body – HTTP 200)

Field	Type	Description
totalRows	integer	Total data rows parsed from the Excel sheet (header row excluded)
successCount	integer	Number of rows that passed all validations and were saved
errorCount	integer	Number of rows that failed validation; records are not saved if this is > 0
message	string	Human-readable summary (e.g. "Upload completed")
totalRecords	long	Equals successCount when no errors
data	array<object>	List of successfully parsed row objects (field names vary by case type)
errors	array<RowError>	Details of rows that failed validation; empty array when no errors

ROWERROR OBJECT

Field	Type	Description
rowNumber	integer	1-based row number in the Excel sheet (row 1 = header, so first data row = 2)
errors	array<string>	Flat list of all validation messages for the row
columnErrors	object<string, array<string>>	Map of column name → list of error messages for that column. null when no column-specific errors.

EXAMPLE – PARTIAL SUCCESS (ROW ERRORS PRESENT)

```
{
  "totalRows": 10,
  "successCount": 0,
  "errorCount": 2,
  "message": "Upload completed",
  "totalRecords": 0,
  "data": [],
  "errors": [
    {
      "rowNumber": 3,
      "errors": ["GSTIN is required", "Zone is required"],
      "columnErrors": {
        "gstin": ["GSTIN is required"],
        "zone": ["Zone is required"]
      }
    },
    {
      "rowNumber": 7,
      "errors": ["Duplicate record exists for GSTIN with OPEN status"],
      "columnErrors": {
        "gstin": ["Duplicate record exists for GSTIN with OPEN status"]
      }
    }
  ]
}
```

EXAMPLE – ALL ROWS VALID (DATA SAVED)

```
{
  "totalRows": 5,
  "successCount": 5,
  "errorCount": 0,
  "message": "Upload completed",
  "totalRecords": 5,
  "data": [ /* array of saved record objects */ ],
  "errors": []
}
```

3.2 ErrorResponse (Upload-Level Failure Body – HTTP 4xx / 5xx)

Returned when the upload fails *before* row processing (e.g. invalid file format, file too large, authentication failure).

Field	Type	Description
error	string	Short error type (e.g. "Bad Request")
errorCode	string	Machine-readable code — see Section 13
message	string	Human-readable error description
status	integer	HTTP status code
path	string	Request URI
traceId	string	Correlation ID for log tracing
timestamp	string (ISO-8601)	Server timestamp
errors	array<string> null	Additional field-level messages (present for validation exceptions)

EXAMPLE – FILE TOO LARGE

```
{
  "error": "Bad Request",
  "errorCode": "errors.file.toolarge",
  "message": "Uploaded file exceeds maximum allowed size",
  "status": 400,
  "path": "/fakeTaxpayer/upload",
  "traceId": "abc123",
  "timestamp": "2026-08-12T10:30:00",
  "errors": null
}
```

4. Fake Taxpayer Identification — Upload

POST /fakeTaxpayer/upload

Uploads an Excel file containing Fake Taxpayer Identification case records.

REQUEST — MULTIPART/FORM-DATA FIELDS

Field	Type	Required	Description
file	File	Required	Excel workbook (.xlsx / .xls , max 10 MB)
caseType	string	Required	Must be "Fake Taxpayer Identification"
subCategory	string	Required	Sub-category of the case (e.g. "Fake Registration")
financialYear	string	Required	Format YYYY-YY (e.g. "2024-25")
createdBy	string	Auto	Injected from the authenticated user's display name

REQUIRED EXCEL COLUMNS

Column Header	Type	Validation
GSTIN	string	Required — exactly 15 uppercase alphanumeric chars <code>^[A-Z0-9]{15}\$</code>
Trade Name	string	Required
Zone	string	Required — must match master zone list
Unit	string	Required — must match master unit list
Circle	string	Required — must match master circle list
Authority	string	Required
Registration Status	Auto	Required
Other columns	various	Optional — stored as-is

DUPLICATE CHECK

Upload is rejected if the **filename** already exists in the system.
Row is rejected if GSTIN + caseType + subCategory + financialYear already exists with status ≠ CLOSED.

SUCCESS RESPONSE (HTTP 200)

Returns [UploadResponse](#). The data array contains objects with all mapped case fields.

5. Non-Filer Cases — Upload

POST /nonfilercases/upload

Uploads Non-Filer Case records. caseType and subCategory are auto-set to "Non-Filer Cases".

REQUEST — MULTIPART/FORM-DATA FIELDS

Field	Type	Required	Description
file	File	Required	Excel workbook (.xlsx / .xls , max 10 MB)
financialYear	string	Required	Format YYYY-YY (e.g. "2024-25")
createdBy	string	Auto	Injected from authenticated user
caseType	string	Auto	Server auto-sets to "Non-Filer Cases"
subCategory	string	Auto	Server auto-sets to "Non-Filer Cases"

REQUIRED EXCEL COLUMNS

Column Header	Type	Validation
GSTIN	string	Required — <code>^[A-Z0-9]{15}\$</code>
Trade Name	string	Required
Zone	string	Required — master-validated
Unit	string	Required — master-validated
Circle	string	Required — master-validated
Other columns	various	Optional

DUPLICATE CHECK

Upload rejected if filename already exists.
Row rejected if GSTIN + caseType + financialYear already exists with status ≠ CLOSED.

SUCCESS RESPONSE (HTTP 200)

Returns [UploadResponse](#).

6. Revenue Intelligence Report — Upload

POST /revenueintelligence/upload

Uploads Revenue Intelligence Report case records. Both caseType and subCategory are validated server-side before row processing.

REQUEST — MULTIPART/FORM-DATA FIELDS

Field	Type	Required	Description
file	File	Required	Excel workbook (.xlsx / .xls , max 10 MB)
caseType	string	Required	Must be "Revenue Intelligence Report"
subCategory	string	Required	Sub-category value (upload-level validation — <code>BadRequestException</code> thrown if blank)
financialYear	string	Required	Format YYYY-YY
createdBy	string	Auto	Injected from authenticated user

REQUIRED EXCEL COLUMNS

Column Header	Type	Validation
GSTIN	string	Required — <code>^[A-Z0-9]{15}\$</code>
Trade Name	string	Required
Zone	string	Required — master-validated

Column Header	Type	Validation
Unit	string	Required — master-validated
Circle	string	Required — master-validated
Authority	string	Required
Other columns	various	Optional

UPLOAD-LEVEL VALIDATION (THROWS HTTP 400 BEFORE ROW PROCESSING)

If caseType is blank → `BadRequestException("Case Type is required")`
If subCategory is blank → `BadRequestException("Sub Category is required")`

DUPLICATE CHECK

Upload rejected if filename already exists.
Row rejected if GSTIN + caseType + subCategory + financialYear already exists with status ≠ CLOSED.

SUCCESS RESPONSE (HTTP 200)

Returns [UploadResponse](#).

7. Other Departmental Reports — Upload

POST /otherDeptReports/upload

Uploads Other Departmental Report case records. Duplicate check logic varies depending on whether subCategory is provided.

REQUEST — MULTIPART/FORM-DATA FIELDS

Field	Type	Required	Description
file	File	Required	Excel workbook (.xlsx / .xls , max 10 MB)
caseType	string	Required	Must be "Other Departmental Reports"
subCategory	string	Required	Sub-category of the report
financialYear	string	Required	Format YYYY-YY
createdBy	string	Auto	Injected from authenticated user

REQUIRED EXCEL COLUMNS

Column Header	Type	Validation
GSTIN	string	Required — <code>^[A-Z0-9]{15}\$</code>
Zone	string	Required — master-validated
Unit	string	Required — master-validated
Circle	string	Required — master-validated
Potential IGST / CGST / SGST / Total	decimal	Optional — numeric format validated
Other columns	various	Optional

DUPLICATE CHECK

Upload rejected if filename already exists.
Row rejected if:

- With subCategory: GSTIN + caseType + subCategory + financialYear with status ≠ CLOSED
- Without subCategory: GSTIN + caseType + financialYear with status ≠ CLOSED

SUCCESS RESPONSE (HTTP 200)

Returns [UploadResponse](#).

8. IGST Reversal — Upload

POST /isgtReversal/upload

Uploads IGST Reversal case records.

REQUEST — MULTIPART/FORM-DATA FIELDS

Field	Type	Required	Description
file	File	Required	Excel workbook (.xlsx / .xls , max 10 MB)
caseType	string	Required	Must be "IGST Reversal"
subCategory	string	Required	Sub-category value
financialYear	string	Required	Format YYYY-YY
createdBy	string	Auto	Injected from authenticated user

REQUIRED EXCEL COLUMNS

Column Header	Type	Validation
GSTIN	string	Required — <code>^[A-Z0-9]{15}\$</code>
Trade Name	string	Required
Zone	string	Required — master-validated
Unit	string	Required — master-validated
Circle	string	Required — master-validated
Potential Reversal Auto	decimal	Optional — numeric format validated
Potential Reversal Officer	decimal	Optional — numeric format validated
IGST Reversed Auto	decimal	Optional — numeric format validated
IGST Reversed Officer	decimal	Optional — numeric format validated

DUPLICATE CHECK

Upload rejected if filename already exists.
Row rejected if GSTIN + caseType + financialYear already exists with status ≠ CLOSED.

SUCCESS RESPONSE (HTTP 200)

Returns [UploadResponse](#).

9. Audit Cases — Upload

POST /auditCases/upload

Uploads Audit Case records.

REQUEST — MULTIPART/FORM-DATA FIELDS

Field	Type	Required	Description
file	File	Required	Excel workbook (.xlsx / .xls , max 10 MB)
caseType	string	Required	Must be "Audit Cases"
subCategory	string	Required	Sub-category value
financialYear	string	Required	Format YYYY-YY
createdBy	string	Auto	Injected from authenticated user

REQUIRED EXCEL COLUMNS

Column Header	Type	Validation
GSTIN	string	Required — <code>^[A-Z0-9]{15}\$</code>
Trade Name	string	Required
Zone	string	Required — master-validated
Unit	string	Required — master-validated
Circle	string	Required — master-validated

Column Header	Type	Validation
Potential Officer	decimal	Optional — numeric format validated
Other columns	various	Optional

DUPLICATE CHECK

Upload rejected if filename already exists.
 Row rejected if GSTIN + caseType + subCategory + financialYear already exists with status ≠ CLOSED.

SUCCESS RESPONSE (HTTP 200)

Returns [UploadResponse](#).

10. Individual Targets – SGST Collection — Upload

POST

/individual-targets-sgst-upload/upload

Uploads monthly SGST collection targets per Zone / Unit / Circle. *No caseType or subCategory parameters required* — these are fixed for this endpoint.

REQUEST — MULTIPART/FORM-DATA FIELDS

Field	Type	Required	Description
file	File	Required	Excel workbook (.xlsx / .xls , max 10 MB)
createdBy	string	Auto	Injected from authenticated user

REQUIRED EXCEL COLUMNS

Column Header	Type	Validation
Zone	string	Required
Unit	string	Required
Circle	string	Required
Target	decimal	Required — numeric value
Financial Year / FY	string	Required — format YYYY-YY
Month	string	Required — e.g. "April" , "May"

DUPLICATE CHECK

Row rejected if **Financial Year + Month** combination already exists in the database.
 Error message: "Record already exists for Financial Year '...' and Month '...'"

SUCCESS RESPONSE (HTTP 200)

Returns [UploadResponse](#). The data array contains objects with fields: zone, unit, circle, target, financialYear, month.

EXAMPLE SUCCESS RESPONSE

```

{
  "totalRows": 3,
  "successCount": 3,
  "errorCount": 0,
  "message": "Upload completed",
  "totalRecords": 3,
  "data": [
    { "zone": "Guwahati", "unit": "Unit-1", "circle": "Circle-A", "target": 1250000.00, "financialYear": "2024-25", "month": "Ap
  ],
  "errors": []
}

```

11. Individual Targets – Non-GST Collection — Upload

POST

/individual-targets-non-gst-upload/upload

Uploads monthly Non-GST collection targets per Zone / Unit / Circle. Identical structure and validation rules to the SGST Collection endpoint.

REQUEST — MULTIPART/FORM-DATA FIELDS

Field	Type	Required	Description
file	File	Required	Excel workbook (.xlsx / .xls , max 10 MB)
createdBy	string	Auto	Injected from authenticated user

REQUIRED EXCEL COLUMNS

Column Header	Type	Validation
Zone	string	Required
Unit	string	Required
Circle	string	Required
Target	decimal	Required — numeric value
Financial Year / FY	string	Required — format YYYY-YY
Month	string	Required

DUPLICATE CHECK

Row rejected if Financial Year + Month already exists.

SUCCESS RESPONSE (HTTP 200)

Returns [UploadResponse](#).

12. Template Download Endpoints

Each case type provides a template endpoint that returns a pre-formatted .xlsx file with the correct column headers.

REQUEST — GET (NO BODY REQUIRED)

All template endpoints require a valid Bearer JWT token.

Case Type	Endpoint	Template Filename
Fake Taxpayer Identification	GET /fakeTaxpayer/download-template	fake_taxpayer_template.xlsx
Non-Filer Cases	GET /nonfilercases/download-template	non_filer_cases_template.xlsx
Revenue Intelligence Report	GET /revenueintelligence/download-template	revenue_intelligence_template.xlsx
Other Departmental Reports	GET /otherDeptReports/download-template	other_dept_reports_template.xlsx
IGST Reversal	GET /isgtReversal/download-template	igst_reversal_template.xlsx
Audit Cases	GET /auditCases/download-template	audit_cases_template.xlsx
Individual Targets – SGST Collection	GET /individual-targets-sgst-upload/download-template	sgst_targets_template.xlsx
Individual Targets – Non-GST Collection	GET /individual-targets-non-gst-upload/download-template	non_gst_targets_template.xlsx

RESPONSE HEADERS

Header	Value
Content-Disposition	attachment; filename="<template_filename>"
Content-Type	application/vnd.openxmlformats-officedocument.spreadsheetml.sheet

13. Error & Exception Codes

13.1 HTTP Status Codes

HTTP Status	When Returned
200 OK	Request processed. Inspect <code>errorCount</code> — data only saved when <code>errorCount</code> <code>≡</code> 0

HTTP Status	When Returned
400 Bad Request	Upload-level failure: invalid file format, missing required parameters, file too large, corrupt workbook, or missing mandatory form fields (caseType / subCategory for Revenue Intelligence)
401 Unauthorized	Missing, expired, or invalid JWT token
403 Forbidden	Authenticated user lacks permission to upload for the requested jurisdiction
409 Conflict	Data integrity violation (e.g., duplicate filename detected at the database level)
500 Internal Server Error	Unexpected server error — check server logs with the <code>traceId</code> from the response

13.2 API Error Codes (errorCode field)

Error Code	Message Key	Trigger Scenario
<code>errors.validation.failed</code>	<code>VALIDATION_FAILED</code>	Bean validation failure on request parameters
<code>errors.request.bodyInvalid</code>	<code>REQUEST_BODY_INVALID</code>	Malformed request body / unreadable multipart form
<code>errors.file.upload</code>	<code>FILE_UPLOAD_ERROR</code>	<code>ExcelUploadException</code> : workbook cannot be opened or parsed, unsupported format
<code>errors.file.tooLarge</code>	<code>FILE_TOO_LARGE</code>	File exceeds 10 MB (Spring <code>MaxUploadSizeExceededException</code>)
<code>errors.common.badRequest</code>	<code>BAD_REQUEST</code>	<code>BadRequestException</code> : missing caseType / subCategory (Revenue Intelligence), invalid arguments, unrecoverable parse error during row processing
<code>errors.resource.notFound</code>	<code>RESOURCE_NOT_FOUND</code>	Requested resource (e.g. template) not found
<code>errors.auth.required</code>	<code>AUTH_REQUIRED</code>	No / expired JWT token
<code>errors.auth.accessDenied</code>	<code>ACCESS_DENIED</code>	Insufficient role or jurisdiction
<code>errors.common.conflict</code>	<code>CONFLICT</code>	Database constraint violation (duplicate filename at DB level)
<code>errors.common.internal</code>	<code>INTERNAL_ERROR</code>	Unhandled server-side error

13.3 Row-Level Validation Error Messages

These appear inside `UploadResponse.errors[].errors` (HTTP 200 response):

Field	Error Message	Affected Case Types
<code>gstIn</code>	"GSTIN is required"	All GSTIN-based case types
<code>gstIn</code>	"Invalid GSTIN format" (must match <code>^[A-Z0-9]{15}\$</code>)	All GSTIN-based case types
<code>gstIn</code>	"Duplicate record exists for GSTIN with same CaseType and FinancialYear in OPEN status"	Non-Filer Cases, Revenue Intelligence, Other Dept Reports, IGST Reversal
<code>gstIn</code>	"Duplicate record exists for GSTIN with OPEN status"	Audit Cases, Fake Taxpayer
<code>tradeName</code>	"Trade Name is required"	All except Other Dept Reports
<code>zone</code>	"Zone is required"	All
<code>unit</code>	"Unit is required"	All
<code>circle</code>	"Circle is required"	All
<code>authority</code>	"authority is required"	Fake Taxpayer, Revenue Intelligence
<code>registrationStatus</code>	"registration status is required"	Fake Taxpayer
<code>target</code>	"Target is required"	SGST & Non-GST Collection
<code>financialYear</code>	"Financial Year is required"	SGST & Non-GST Collection
<code>month</code>	"Month is required"	SGST & Non-GST Collection
<code>financialYear</code> + <code>month</code>	"Record already exists for Financial Year '...' and Month '...'"	SGST & Non-GST Collection

13.4 Upload-Level Error Examples

These result in HTTP 4xx responses (not HTTP 200) — data is never processed.

MISSING / INVALID JWT (HTTP 401)

```
{
  "error":      "Unauthorized",
  "errorCode":  "errors.auth.required",
  "message":    "Full authentication is required to access this resource",
  "status":     401,
  "path":       "/fakeTaxpayer/upload",
  "timestamp":  "2026-08-12T10:00:00",
  "errors":     null
}
```

FILE TOO LARGE (HTTP 400)

```
{
  "error":      "Bad Request",
  "errorCode":  "errors.file.toolarge",
  "message":    "Uploaded file exceeds maximum allowed size",
  "status":     400,
  "path":       "/nonfilercases/upload",
  "timestamp":  "2026-08-12T10:01:00",
  "errors":     null
}
```

CORRUPT / UNREADABLE EXCEL (HTTP 400)

```
{
  "error":      "Bad Request",
  "errorCode":  "errors.common.badRequest",
  "message":    "Upload failed: Unable to open Excel workbook",
  "status":     400,
  "path":       "/auditCases/upload",
  "timestamp":  "2026-08-12T10:02:00",
  "errors":     null
}
```

MISSING SUBCATEGORY FOR REVENUE INTELLIGENCE (HTTP 400)

```
{
  "error":      "Bad Request",
  "errorCode":  "errors.common.badRequest",
  "message":    "Sub Category is required",
  "status":     400,
  "path":       "/revenueintelligence/upload",
  "timestamp":  "2026-08-12T10:03:00",
  "errors":     null
}
```

14. Common Validation Rules

Rule	Detail
File format	Only .xlsx (OOXML) and .xls (BIFF8) formats accepted. Other extensions return HTTP 400.
File size	Maximum 10 MB. Larger files return HTTP 400 with error code errors.file.toolarge .
Empty file / no data rows	Returns HTTP 200 with totalRows: 0 , successCount: 0 , empty errors array. No records are saved.
Header row	Row 0 (first row) is always treated as the header. Column headers are normalised: trimmed, lowercased, non-alphanumeric characters removed before matching.
Empty rows	Completely blank rows are skipped silently and not counted in totalRows .
GSTIN format	Must match regex ^[A-Z0-9]{15}\$ — exactly 15 uppercase alphanumeric characters. Lowercase letters and special characters are invalid.
Numeric fields	Fields typed as decimal / BigDecimal / Double accept numeric cell values. Text that cannot be parsed to a number is treated as null.

Rule	Detail
Filename uniqueness	For GSTIN-based case types, the uploaded filename must be unique. A duplicate filename causes the entire upload to be rejected (HTTP 409 or row-level error depending on implementation).
GSTIN duplicate detection	A GSTIN is considered duplicate only when the matching record has status <code>≠ CLOSED</code> . Closed records allow re-upload of the same GSTIN.
Transactional behaviour	If <i>any</i> row fails validation, no rows are saved . The entire upload is atomic — all rows must pass for data to be persisted.
Zone / Unit / Circle	Values are validated against the jurisdiction master via <code>ZoneUnitCircleValidationService</code> . Values not in the master list generate a row-level error.
Financial Year format	Expected format: <code>YYYY-YY</code> (e.g. <code>2024-25</code>). The field name in the template may be <code>Financial Year</code> or <code>FY</code> — both are recognised.

Quick Reference – All Upload Endpoints

Case Type	Endpoint	Key Form Parameters	GSTIN Required
Fake Taxpayer Identification	<code>POST /fakeTaxpayer/upload</code>	<code>file</code> , <code>caseType</code> , <code>subCategory</code> , <code>financialYear</code>	Yes
Non-Filer Cases	<code>POST /nonfilercases/upload</code>	<code>file</code> , <code>financialYear</code>	Yes
Revenue Intelligence Report	<code>POST /revenueintelligence/upload</code>	<code>file</code> , <code>caseType</code> , <code>subCategory*</code> , <code>financialYear</code>	Yes
Other Departmental Reports	<code>POST /otherDeptReports/upload</code>	<code>file</code> , <code>caseType</code> , <code>subCategory</code> , <code>financialYear</code>	Yes
IGST Reversal	<code>POST /isgtReversal/upload</code>	<code>file</code> , <code>caseType</code> , <code>subCategory</code> , <code>financialYear</code>	Yes
Audit Cases	<code>POST /auditCases/upload</code>	<code>file</code> , <code>caseType</code> , <code>subCategory</code> , <code>financialYear</code>	Yes
Individual Targets – SGST Collection	<code>POST /individual-targets-sgst-upload/upload</code>	<code>file</code>	No (zone/unit/circle/target/month)
Individual Targets – Non-GST Collection	<code>POST /individual-targets-non-gst-upload/upload</code>	<code>file</code>	No (zone/unit/circle/target/month)

* `subCategory` for Revenue Intelligence is validated server-side before row processing and returns HTTP 400 if blank.